

ARCHITECTURE BRIEFING

Zero Trust Network Segmentation

Leveraging CrowdStrike Falcon Agents to Seed, Discover, and Accelerate Illumio
Application Dependency Mapping.

Concept & Integration Map

Discovering how host telemetry and cloud-to-cloud security orchestrations construct robust microsegmentation policies.

System Architecture Flow

Telemetry to Enforcement Flow

The northbound-southbound relationship enables single-agent management across enterprise workloads:

- **Northbound Telemetry:** CrowdStrike sensors track process-to-process interactions and route them safely to CrowdStrike Cloud.
- **PCE Ingestion:** Illumio's connector consumes FDR/CFIS streams, translating active endpoints into map nodes.
- **Southbound Deployment:** Policy rules are generated and deployed to agents to command native host firewalls.



600 × 400

Rich Endpoint Telemetry



Network Connections

Captures dynamic source/destination IPs, protocols, active TCP/UDP ports, and directional flows (inbound vs. outbound) directly at host-level execution.



Process Lineage

Maps child-to-parent processes, pinpointing exact execution files initiating sessions to ensure segment enforcement targets trusted binaries.



Asset Metadata

Extracts localized operating system information, domain context, cloud platform IDs, and tags to drive policy categorization inside Illumio.

Map Mapping in Illumio

Dependency Visualization

Illumio aggregates the ingested Falcon agent connectivity metadata to visualize actual application structures. This immediately identifies east-west server communications, hidden APIs, and anomalous traffic flows without network probes.

Targeted Ring-Fencing

Once flows are validated, Illumio automatically recommends specific microsegmentation policies. For instance, creating targeted rules allowing SAP Servers to interact with SQL instances exclusively over authorized database ports.

 Platform Feature Mapping

Core Capability	CrowdStrike Falcon	Illumio Core Platform
Host-Level Connection Capture	✔ Primary Agent	✔ Supported (PCE/VEN)
App Dependency Mapping	⚠ Basic Inventory	✔ Advanced Visibility Map
Microsegmentation Policy Engine	✖ No Native Engine	✔ Advanced Modeling & Sim
Zero Trust Enforcement	✔ OS Firewall Management	✔ VEN Agent (Windows/Linux/OS)
Interactive Policy Simulation	✖ No Simulation	✔ Full Dry-Run Simulation

Standard Integration Workflow

1

Collect Data

Falcon agents capture system events and direct connectivity metadata directly from the endpoints.

2

Ingest & Correlate

CrowdStrike Cloud pipes logs to FDR, allowing Illumio's connector to import normalized flow logs.

3

Model & Design

Illumio PCE auto-builds maps and simulates Zero Trust rules to ensure business continuity.

4

Enforce Policies

Secure segmentation is applied locally via Illumio VEN agents or CrowdStrike native firewall rules.

Deployment Velocity Impact

0

SPAN Ports or Network TAPs Required

Reusing Endpoint Telemetry

Traditional application mapping relies heavily on tedious span-port replication, network-level routing re-configurations, or hardware-based appliance additions. Re-using current CrowdStrike Falcon agents eliminates these operations entirely, compressing historical visibility discovery times from several months down to just minutes.

Strategic Integration Value

Agent Consolidation

By leveraging the active Falcon installation for both telemetry detection and operating-system-level firewall configurations (via Illumio Edge/Firewall Management), organizations avoid agent bloat on critical user workstations.

High-Fidelity Intelligence

Ingesting telemetry directly from the host operating system provides process and binary contexts that standard network sniffers miss. This prevents the creation of flawed rules by clearly identifying which executables are active.

☑️ Key Architecture Recommendations

➡️ Leverage Falcon for Seeding Map

Always utilize initial CrowdStrike FDR telemetry to seed mapping phases. It establishes immediate visibility prior to planning microsegmentation boundaries.

➡️ Evaluate OS Limitations Separately

Verify workload profiles. Standard endpoints (Windows/macOS laptops) can leverage unified agent enforcement, while production servers may require dedicated Illumio VENS.

➡️ Audit Licensing Prerequisites

Ensure the CrowdStrike license supports the Falcon Data Replicator (FDR) or CFIS, and Falcon Firewall Management modules are activated prior to building the link.

➡️ Phased Zero Trust Enforcement

Conduct progressive policy testing inside Illumio. Simulate all suggested parameters extensively before deploying restrictive, structural blocking policies.

Questions & Discussion

Achieving host-based Zero Trust through integrated detection and segmentation.

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